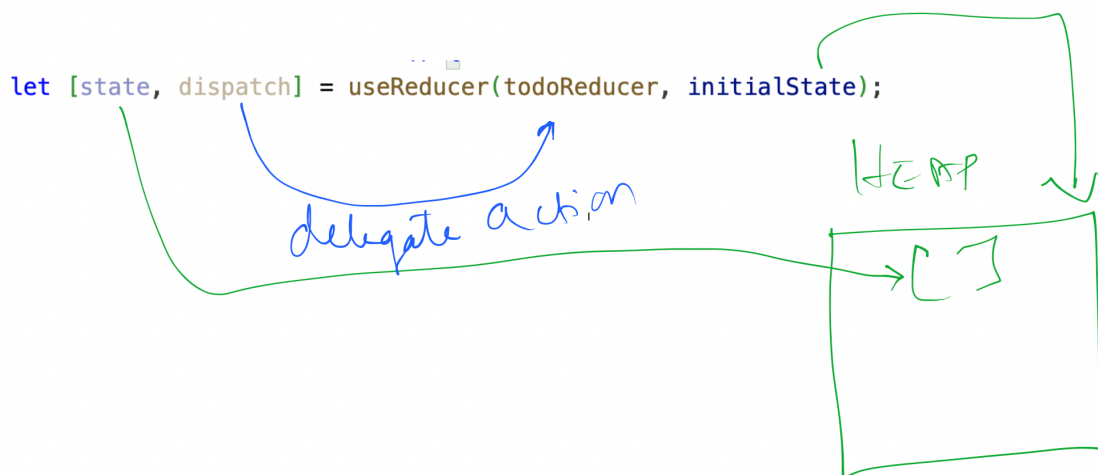
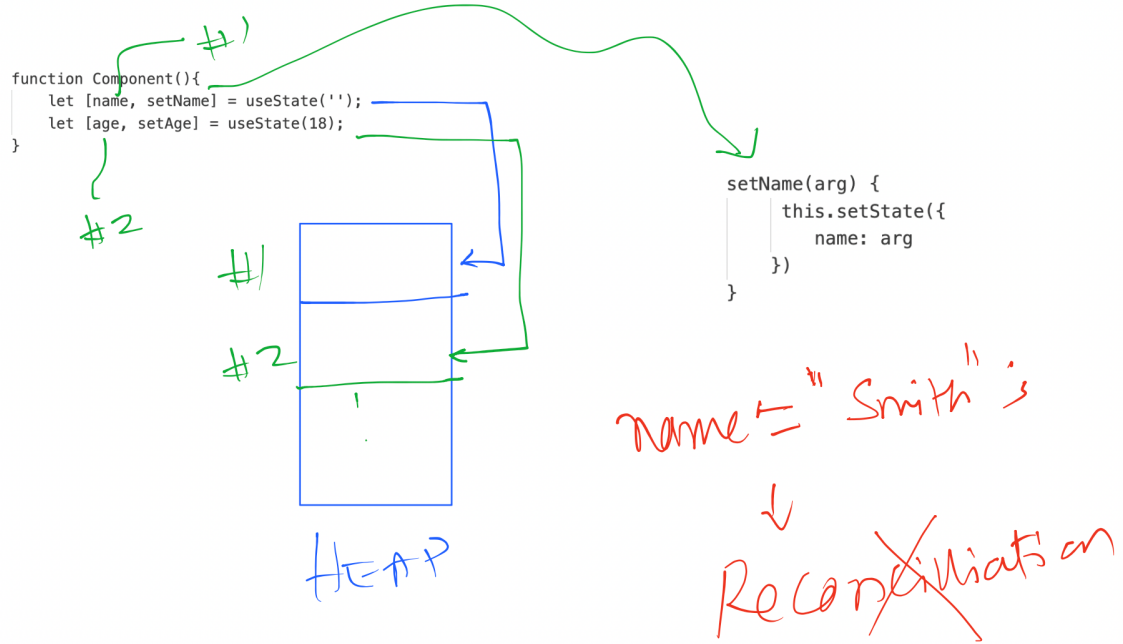




```

export default function TodoList() {
  let [state, dispatch] = useReducer(todoReducer, initialState);
  let [text, setText] = useState('');

  function addTodo() {
    dispatch({type: 'ADD', payload: text})
  }

  return (
    <div>
      <h1>Todo List</h1>
      <input type="text" onChange={(event) => setText(event.target.value)} />
      <button type="button" onClick={() => addTodo()}>Add todo</button>
    </div>
  )
}

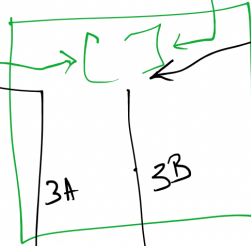
```

```

export default function todoReducer( state, action ) {
  switch(action.type) {
    case 'ADD':
      // avoid using state.push({id: Date.now(), text: action.payload, completed: false});
      // clone and add new payload
      return [...state, {id: Date.now(), text: action.payload, completed: false}]
    case 'TOGGLE':
      // payload will be id
      return state.map(todo => todo.id === action.payload ? {...todo, completed: !todo.completed}: todo)
    case 'REMOVE':
      return state.filter(todo => todo.id !== action.payload);
    default:
      return state;
  }
}

```

HEAP



4

3A

3B

1

3

2